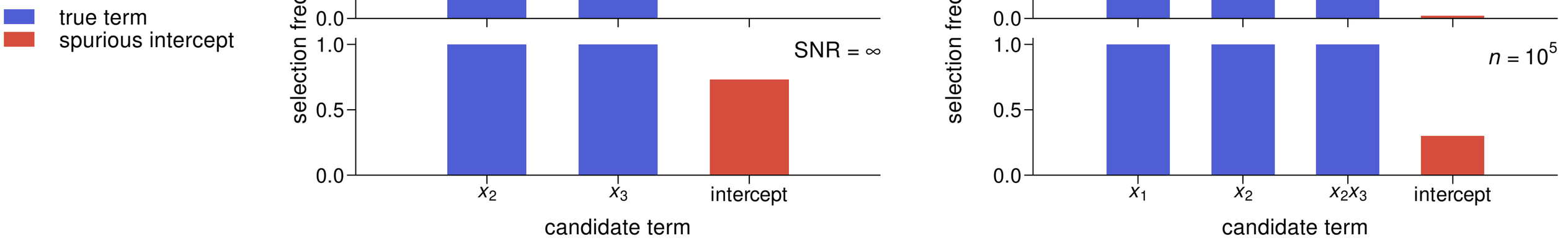
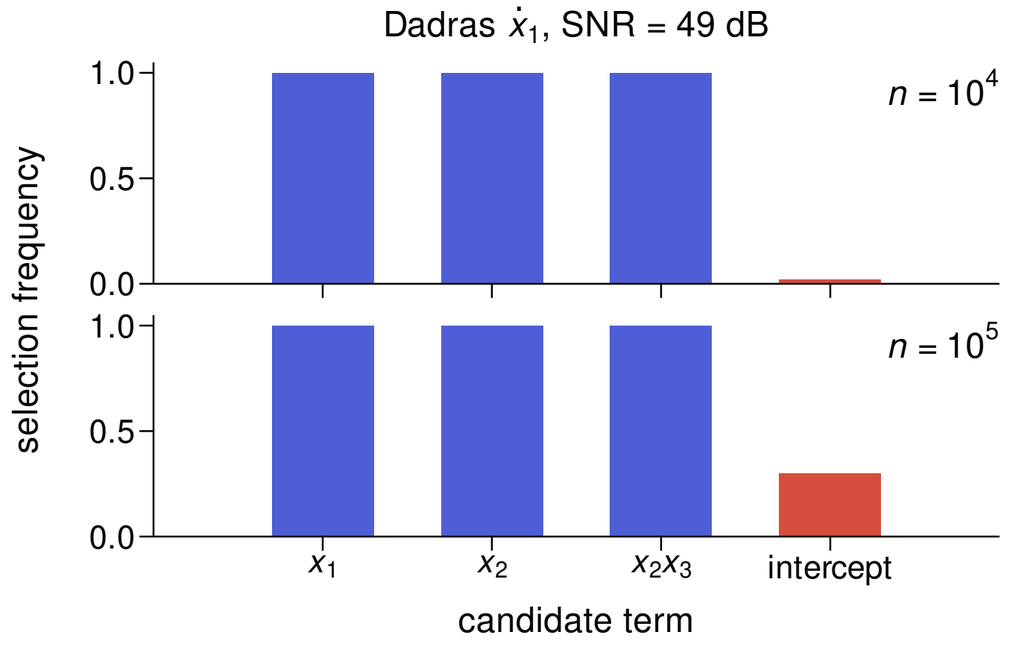
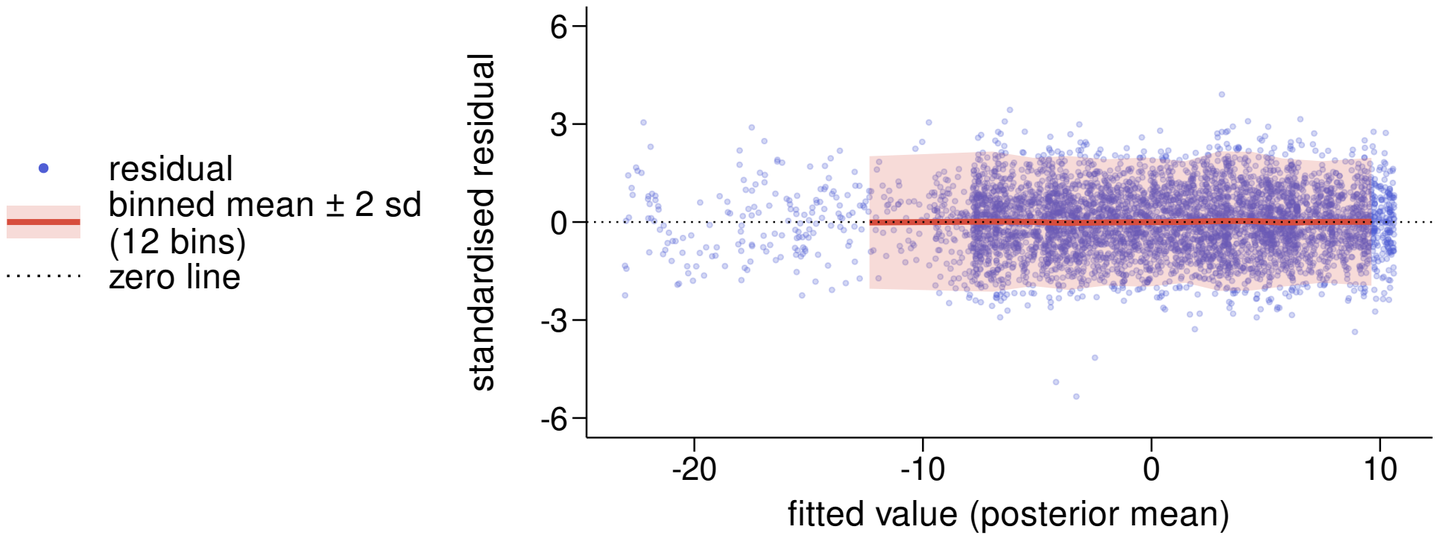
a


true term
spurious intercept
Rössler $\dot{x}_1$, $n = 5000$
SNR = 61 dB
SNR = ∞
selection frequency
1.0
0.5
0.0
x_2
x_3
intercept
candidate term

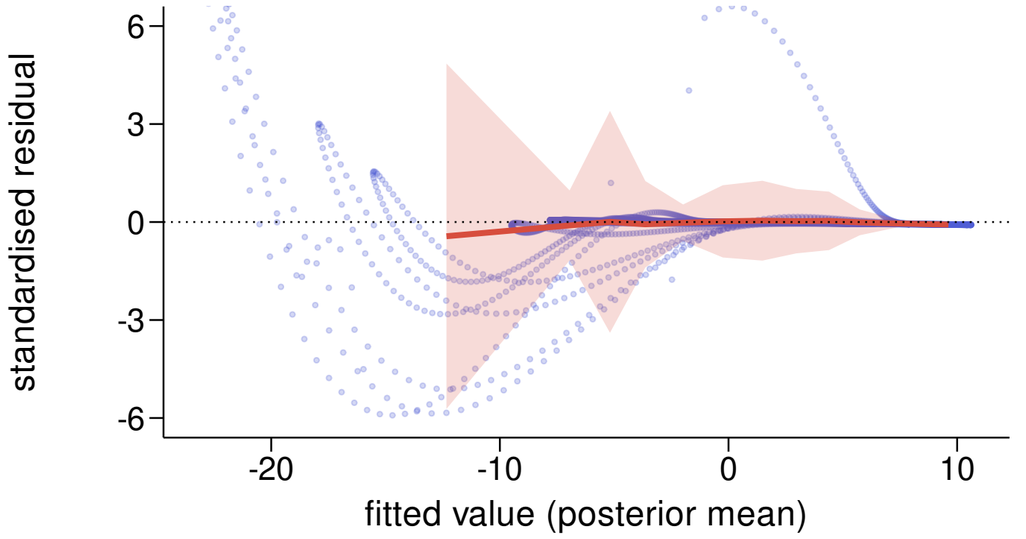
b


Dadras $\dot{x}_1$, SNR = 49 dB
$n = 10^4$
$n = 10^5$
selection frequency
1.0
0.5
0.0
x_1
x_2
x_2x_3
intercept
candidate term

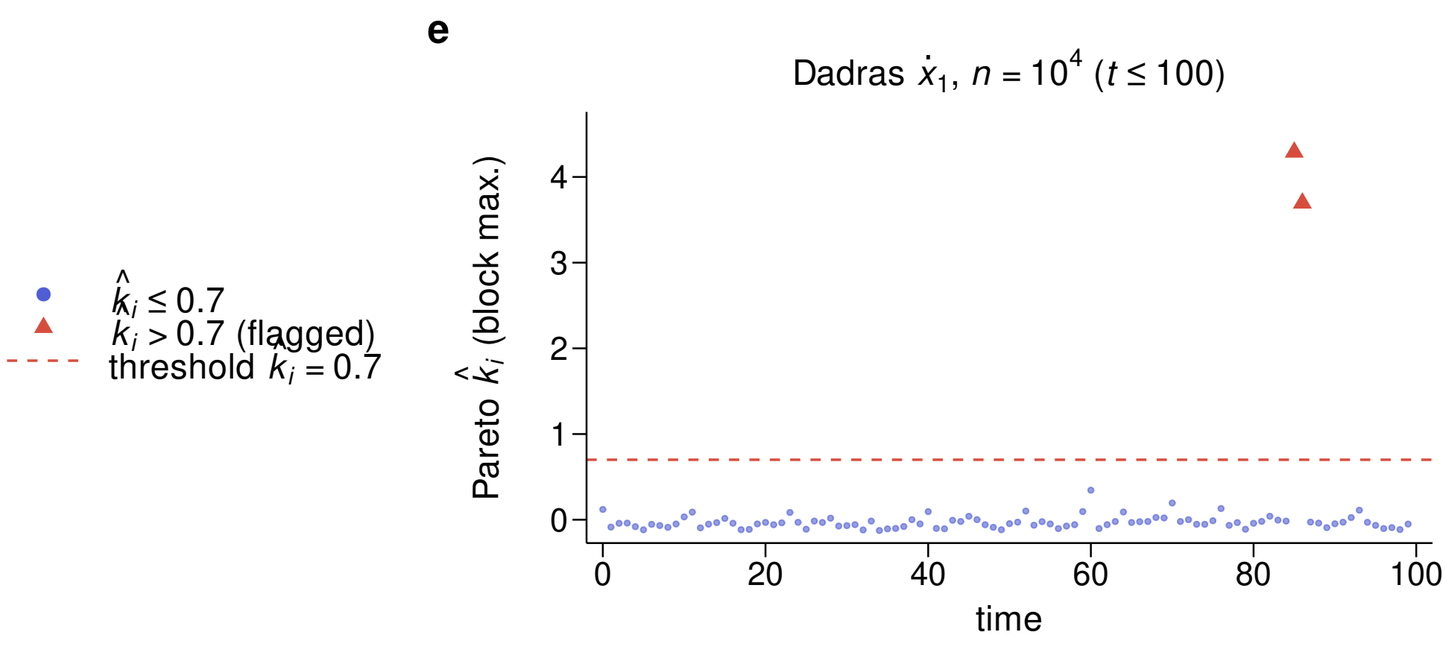
c


residual
binned mean ± 2 sd
(12 bins)
zero line
Rössler $\dot{x}_1$, SNR = 61 dB
standardised residual
6
3
0
-3
-6
-20
-10
0
10
fitted value (posterior mean)

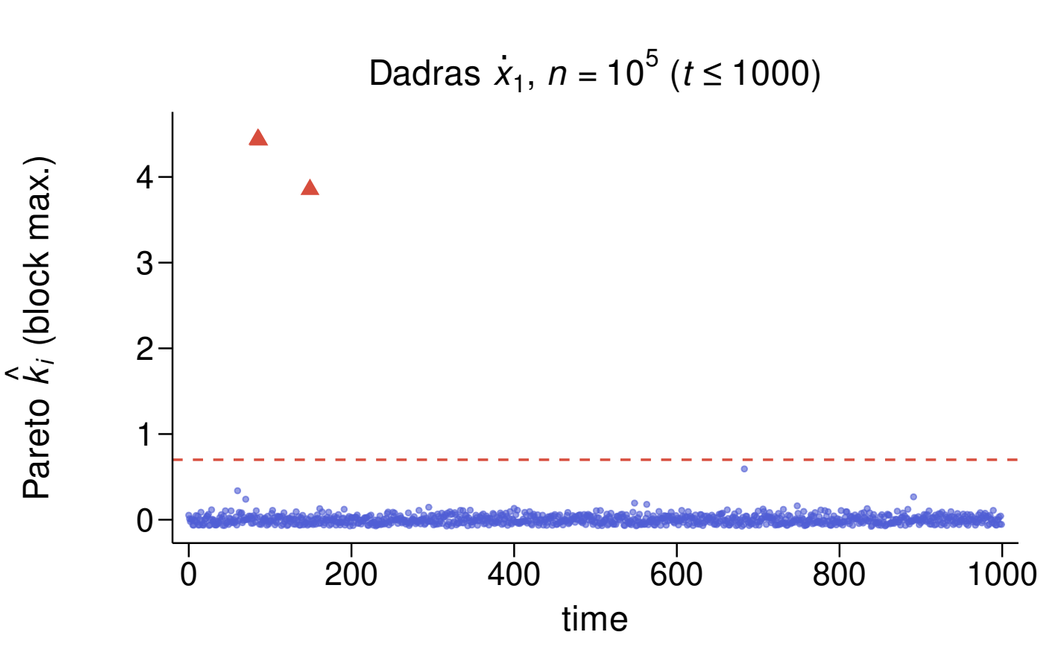
d


Rössler $\dot{x}_1$, SNR = ∞
standardised residual
6
3
0
-3
-6
-20
-10
0
10
fitted value (posterior mean)

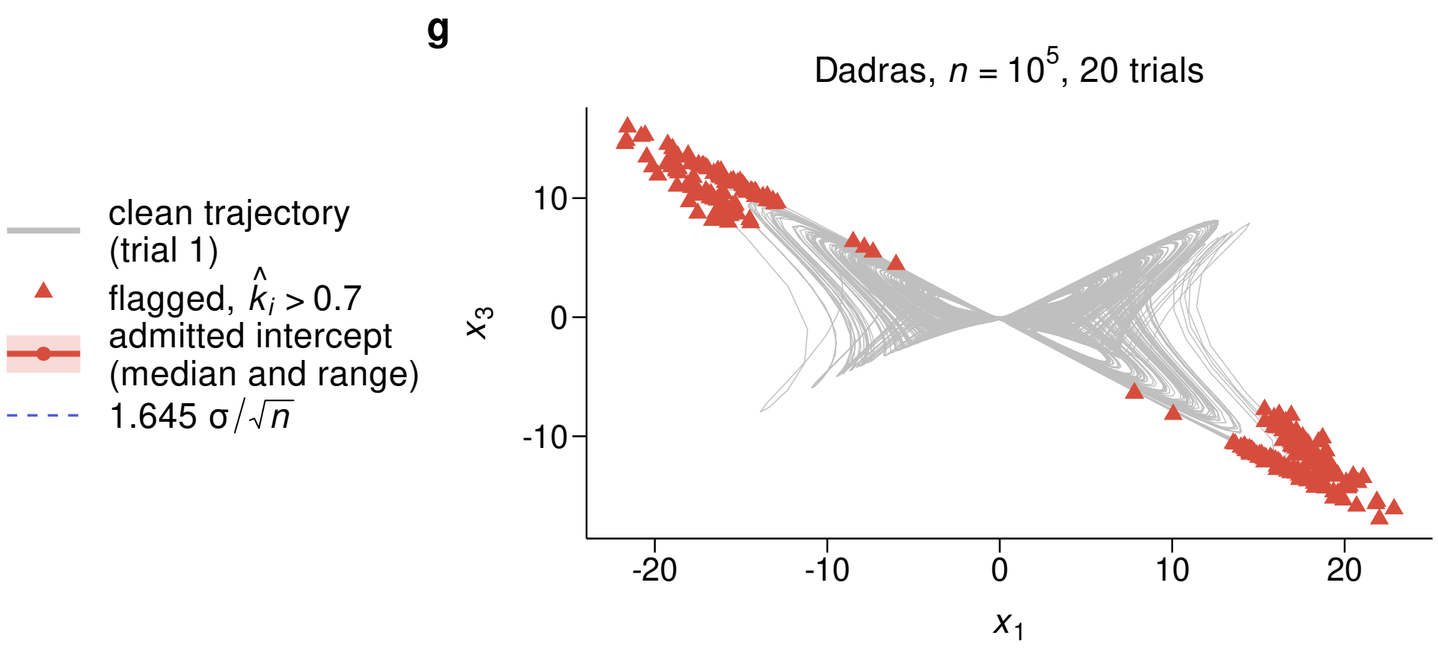
e


$\hat{k}_i \le 0.7$
$\hat{k}_i > 0.7$ (flagged)
threshold $\hat{k}_i = 0.7$
Dadras $\dot{x}_1$, $n = 10^4$ ($t \le 100$)
Pareto $\hat{k}_i$ (block max.)
4
3
2
1
0
0
20
40
60
80
100
time

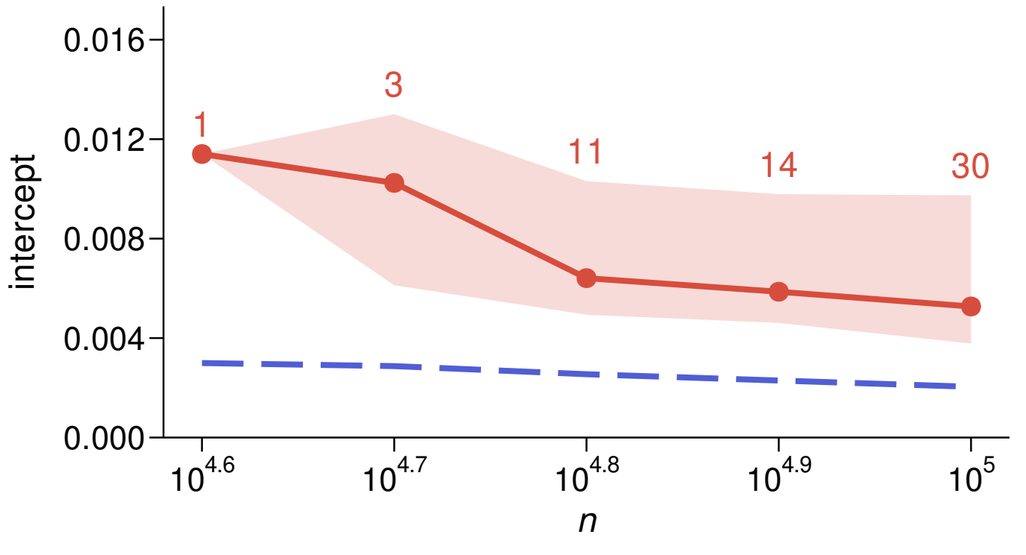
f


Dadras $\dot{x}_1$, $n = 10^5$ ($t \le 1000$)
Pareto $\hat{k}_i$ (block max.)
4
3
2
1
0
0
200
400
600
800
1000
time

g


clean trajectory
(trial 1)
flagged, $\hat{k}_i > 0.7$
admitted intercept
(median and range)
$1.645\ \sigma/\sqrt{n}$
Dadras, $n = 10^5$, 20 trials
x_3
10
0
-10
-20
-10
0
10
20
x_1

h


Dadras $\dot{x}_1$, SNR = 49 dB
intercept
0.016
0.012
0.008
0.004
0.000
1
3
11
14
30
$10^{4.6}$
$10^{4.7}$
$10^{4.8}$
$10^{4.9}$
10^5
n